

Curriculum Vitae

Machine learning researcher with an MSc in Biomedical Engineering (5/5, Excellent, June 2026) from the University of Oulu, specializing in deep learning, computer vision, and multimodal data analysis. Research at the Center for Machine Vision and Signal Analysis (CMVS) encompasses transformer-based architectures for multi-sequence image segmentation, affective computing, and the application of deep learning to bio signal and image data. First author of a manuscript under review at Information Fusion (Elsevier). Brings hands-on experience in multimodal data processing, transformer model design, and reproducible experimentation on high-performance computing infrastructure. Seeking to contribute to computer vision and generative model research in the CMVS environment.

1. Personal Details and Date of the CV

Surname: Mridha

First names: Ramisa Fariha

Researcher ID: ORCID: 0009-0001-5194-586X (orcid.org/0009-0001-5194-586X)

Contact: Oulu, Finland | rf.mridha.13@gmail.com | +358 44 928 9773

LinkedIn: [linkedin.com/in/ramisa-fariha-mridha-6a25a01ab](https://www.linkedin.com/in/ramisa-fariha-mridha-6a25a01ab) **GitHub:** github.com/Ramisa13

Date of CV: 8 July 2026

2. Degrees

- **15 June 2026 (degree conferred 30 June 2026).** Master of Science (Technology), Biomedical Engineering (Computer Science and Engineering), Specialization: Medical Data Analysis. University of Oulu. GPA 4.57/5.0, 137/120 credits. Master's thesis assessed 5/5 (Excellent). Faculty of Information Technology and Electrical Engineering.
- **2023.** Bachelor of Science in Electrical and Electronic Engineering. Rajshahi University of Engineering and Technology (RUET), Bangladesh. GPA 3.55/4.00, First Class.

3. Other Education and Expertise

- **2024-2025.** Special Course in Information Technology 1: Introduction to Computer Vision Methods for Biomedical Images, and Special Course in Information Technology 2: Function and Analysis of the Cardiovascular System. University of Oulu.
- **2024-2026.** Affective Computing (5/5); Machine Vision (4/5); Machine Learning (5/5); Deep Learning (3/5); Biosignal Processing I and II (5/5); Big Data Processing and Applications (5/5). University of Oulu.

4. Language Skills

Native language: Bengali (Bangla)

- English: C1 (self-assessment)
- Finnish: A1 (self-assessment)

5. Current Employment

Available immediately following graduation (June 2026); actively seeking research positions in computer vision, machine learning, and multimodal data analysis.

- **Research career stage:** Stage I, early-stage researcher.
- **2025-present.** Student Faculty Representative, Halloped, ITEE, OYY, University of Oulu.

6. Previous Work Experience

- **Jul-Aug 2025.** Research Intern, Advanced Practical Training, CMVS, University of Oulu. Supervisors: Dr. Md. Ziaul Hoque and Prof. Tapio Seppanen. Systematic evaluation of deep learning architectures for multimodal image segmentation.
- **2023-2024.** Lecturer, Mathematics, Green University of Bangladesh, Dhaka, Bangladesh.
- **2020-2022.** Chief Technical Officer, Radio RUET, Rajshahi University of Engineering and Technology.
- **2022.** Trainee, Bheramara 410 MW Combined Cycle Power Plant, Khulna, Bangladesh.

8. Research Funding and Grants

- **2025.** ITEE Master's Thesis Scholarship, University of Oulu. Competitive funding for outstanding master's thesis research.

9. Research Output

Manuscripts under review

- Mridha, R. F., et al. Myocardial Infarction Segmentation: Method Benchmarking Using Cardiac Magnetic Resonance Images. Information Fusion, Elsevier (under review, INFFUS-D-26-03942).

Thesis

- Mridha, R. F. (2026). Master's thesis, University of Oulu, assessed 5/5 (Excellent). A novel deep learning approach was developed for multi-sequence cardiac MRI segmentation. (Title withheld pending related journal submission.)

Selected projects

- **EmotiNet:** Comparative study of fine-tuned transformer models for fine-grained emotion recognition from multimodal signals (2024).

- **Diabetic Retinopathy Detection:** Transfer learning with Explainable AI (Grad-CAM) for retinal fundus image classification (2024).
- **Cardiac Image Segmentation with Machine Learning:** Automated segmentation pipeline for cardiac structures from medical images (2023).
- **Fashion-MNIST Classification:** Convolutional neural network benchmark study (2023).

10. Teaching Merits

- 2023-2024. Lecturer in Mathematics, Green University of Bangladesh: undergraduate engineering course delivery.

11. Awards and Honours

- ITEE Master's Thesis Scholarship 2025, University of Oulu.
- 2nd Runner-up, Beetle-weight Robo Rumble, Tech-Mayhem 2023, RUET.

12. Scientific and Societal Impact

- Research on multimodal deep learning and transformer architectures for image and signal analysis contributes to reproducible, open-science practices in computer vision and medical AI.
- Faculty-level student representation through the Halloped programme (2025-present), contributing to ITEE education policy.

13. Other Merits

Technical skills

- **Computer Vision and Generative Models:** Image segmentation, classification, transfer learning, Grad-CAM, transformer architecture, attention mechanisms, multimodal fusion
- **Affective Computing and Biosignals:** Emotion recognition from multimodal data (video, audio, biosignals), Fourier analysis, filtering, cardiovascular signal processing
- **Machine Learning and AI:** Deep Learning (PyTorch), CNN, transformer/LLM-based models, multi-modal learning, systematic benchmarking, loss function design
- **Programming:** Python (NumPy, Pandas, Matplotlib), C/C++, MATLAB
- **Computing and Tools:** CSC Puhti HPC, Git/GitHub, Docker, reproducible research workflows

References

Dr. Md. Ziaul Hoque

Postdoctoral Researcher and University Teacher, ITEE, University of Oulu
Data/ML Scientist, Helsinki University Hospital (HUS), Inborn Group, University of Helsinki
Email: mdziaul.hoque@oulu.fi Tel: +358 50 347 6233

Dr. Anja Keskinarkaus

University Lecturer, ITEE, University of Oulu
Email: anja.keskinarkaus@oulu.fi Tel: +358 40 583 1116